Exploring Topology and Non-Hermiticity with Electrical Circuits

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Why topolectric circuits?

A tight-binding model is mostly about *connectivity* and *coupling strengths*. A circuit network is built from the same two ideas — nodes and links — so we can realize many lattice models directly in hardware.

- The lattice structure becomes the **wiring diagram**, and couplings are set by **component values**.
- Mode shapes are **measurable**: node voltages give the spatial profile of an eigenmode.
- Edge physics becomes **obvious** through localized voltages and impedance peaks [1].

Core bridge

$$H\phi = E\phi \qquad \Longleftrightarrow \qquad I(\omega) = J(\omega)\, V(\omega)$$

[1] Ching Hua Lee et al. "Topolectrical Circuits". In: *Communications Physics* 1.1 (July 2018), p. 39

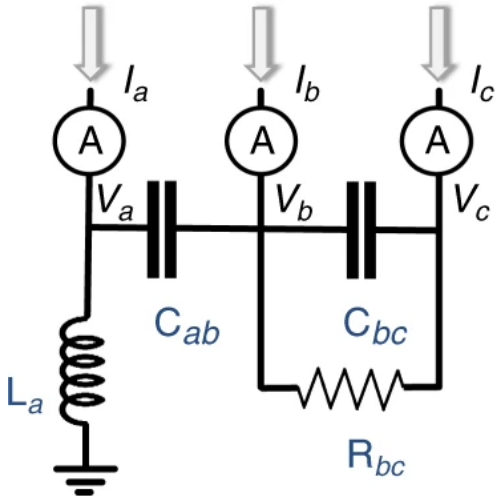
Kirchhoff's law and the grounded circuit Laplacian

At node a, Kirchhoff's current law (KCL) in the admittance language is written as

$$I_a = \sum_i C_{ai}\,(V_a - V_i) + w_a V_a. \tag{1}$$

Stacking all node equations gives the matrix form

$$I = (L + W)\,V := J\,V. \tag{2}$$



Circuit ground

$$W = \begin{pmatrix} (i\omega L_a)^{-1} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Adjacency matrix

$$C = \begin{pmatrix} 0 & i\omega C_{ab} & 0 \\ i\omega C_{ab} & 0 & i\omega C_{bc} + R_{bc}^{-1} \\ 0 & i\omega C_{bc} + R_{bc}^{-1} & 0 \end{pmatrix}$$

Total node conductance

$$D = \begin{pmatrix} i\omega C_{ab} & 0 & 0 \\ 0 & i\omega(C_{ab} + C_{bc}) + R_{bc}^{-1} & 0 \\ 0 & 0 & i\omega C_{bc} + R_{bc}^{-1} \end{pmatrix}$$

Figure: Definition of the grounded circuit Laplacian $J = D - C + W$ and how circuit elements contribute to W, C, and D - adapted from (Lee et al., "Topolectrical Circuits").

Eigenvalue problem of the circuit Laplacian

For an undriven network ($I = 0$), non-trivial voltage patterns exist only when J becomes singular:

$$I = 0 \quad \Longrightarrow \quad J\,V = 0, \qquad \det J = 0. \tag{3}$$

This motivates the eigenvalue problem

$$J\,\psi_n = j_n\,\psi_n, \tag{4}$$

where ψ_n is the circuit mode profile and j_n is its admittance eigenvalue. For periodic lattices one uses a Bloch ansatz,

$$\psi_n(x) = e^{ik\cdot x}\,\phi_n(k), \tag{5}$$

and compares the resulting band structure with a tight-binding problem through the formal identification

$$H_{\mathrm{eff}}(k) \propto i\,J(k,\omega). \tag{6}$$

Two-point impedance and the Green's function

The main experimental observable is the two-point impedance between nodes a and b, defined by driving a current and measuring a voltage drop:

$$Z_{ab} = \frac{V_a - V_b}{I}. \tag{7}$$

Using the inverse Laplacian (the circuit Green's function),

$$V = J^{-1} I = G I, \qquad G = J^{-1}, \tag{8}$$

the impedance can be written directly in terms of Green-function entries:

$$Z_{ab} = G_{aa} + G_{bb} - G_{ab} - G_{ba}. \tag{9}$$

Mode expansion and why impedance peaks appear

In terms of circuit eigenmodes (excluding exact zero-admittance modes),

$$G = \sum_{n,\, j_n \neq 0} \frac{\psi_n \psi_n^\dagger}{j_n}, \tag{10}$$

which gives an intuitive spectral decomposition of the impedance:

Two-Point Impedance

$$Z_{ab} = \sum_{n,\, j_n \neq 0} \frac{|\psi_{n,a} - \psi_{n,b}|^2}{j_n}. \tag{11}$$

So impedance peaks are dominated by modes with small $|j_n|$, and become especially strong when the corresponding mode has weight on the probed nodes (e.g. boundary-localized modes in the SSH circuit).

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SSH chain: dimerization in 1D

The SSH model is a 1D chain with two sites per unit cell (A,B) and alternating couplings [2]: an intracell coupling v and an intercell coupling w. The key point is that swapping v and w changes the topology under open boundaries.



Figure: Dimerized SSH chain with two sublattices and alternating couplings v and w.

[2] W. P. Su, J. R. Schrieffer, and A. J. Heeger. "Solitons in Polyacetylene". In: *Phys. Rev. Lett.* 42 (25 June 1979), pp. 1698–1701. DOI: 10.1103/PhysRevLett.42.1698. URL: https://link.aps.org/doi/10.1103/PhysRevLett.42.1698

SSH Hamiltonian and bulk bands

In real space the tight-binding Hamiltonian is

$$H = \sum_n \left(v\, a_n^\dagger b_n + w\, a_{n+1}^\dagger b_n + \text{h.c.} \right). \tag{12}$$

With periodic boundary conditions, the Bloch Hamiltonian becomes

$$H(k) = \begin{pmatrix} 0 & v + we^{-ik} \\ v + we^{ik} & 0 \end{pmatrix}, \tag{13}$$

giving the dispersion

$$E_\pm(k) = \pm\sqrt{v^2 + w^2 + 2vw\cos k}. \tag{14}$$

The gap closes at $v = w$, which marks the transition point between the two phases [3].

[3] Navketan Batra and Goutam Sheet. "Physics with Coffee and Doughnuts: Understanding the Physics Behind Topological Insulators Through Su-Schrieffer-Heeger Model". In: *Resonance* 25.6 (June 2020), pp. 765–786. ISSN: 0973-712X. DOI: 10.1007/s12045-020-0995-x

Winding number and the phase transition

Write the off-diagonal term as a complex quantity:

$$v + we^{ik} = |v + we^{ik}|\, e^{i\phi(k)}. \tag{15}$$

The winding number counts how $\phi(k)$ winds as k runs across the Brillouin zone:

$$\nu = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{d\phi(k)}{dk}\, dk = \begin{cases} 1, & v < w, \\ 0, & v > w. \end{cases} \tag{16}$$

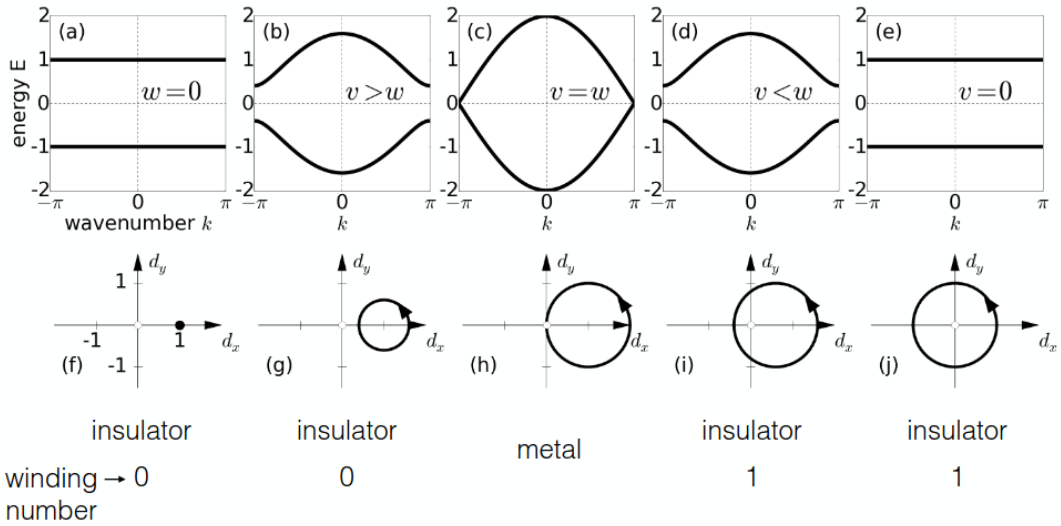


Figure: Bulk bands and the winding picture across parameter regimes (gap closing at $v = w$) - adapted from Batra and Sheet.

Circuit SSH: mapping and LTSpice implementation

Alternating capacitors set the dimerized couplings; each node is grounded by an inductor:

$$C_1,\ C_2 \text{ (alternating links)}, \qquad L \text{ (ground)}.$$

Choose the operating point where the diagonal reactive term cancels:

$$i\omega_0(C_1 + C_2) + \frac{1}{i\omega_0 L} = 0 \qquad \Rightarrow \qquad f_0 = \frac{1}{2\pi\sqrt{L(C_1 + C_2)}}. \tag{17}$$

Near f_0, the phase is mainly controlled by C_1/C_2 (analogue of v/w).

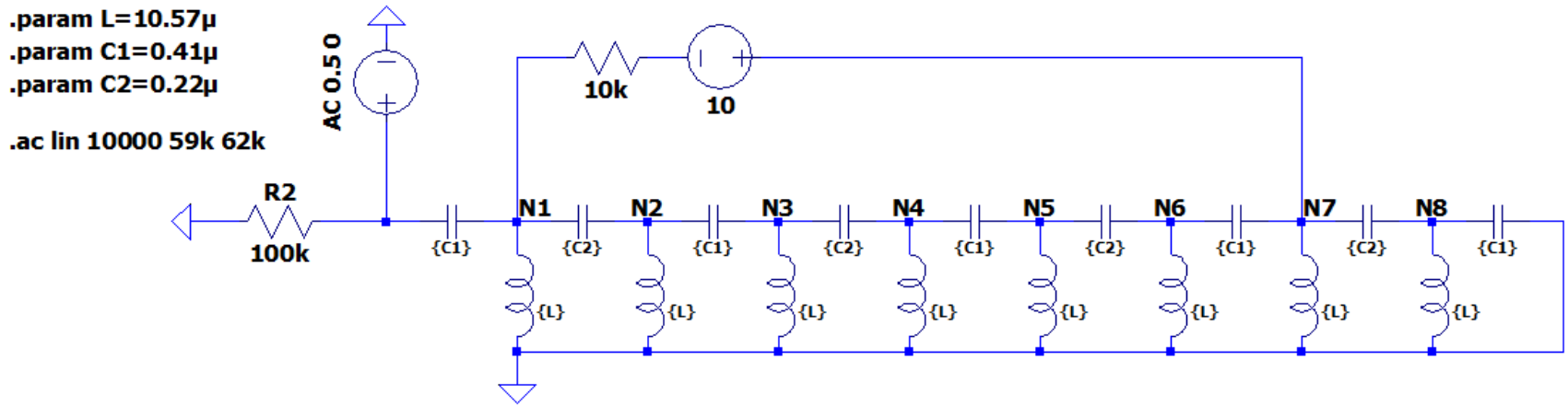


Figure: LTSpice SSH circuit (alternating C_1, C_2; grounding L).

LTSpice results: edge-localized voltage profiles

At the midgap resonance, the node voltages become strongly asymmetric, indicating localization towards one boundary.

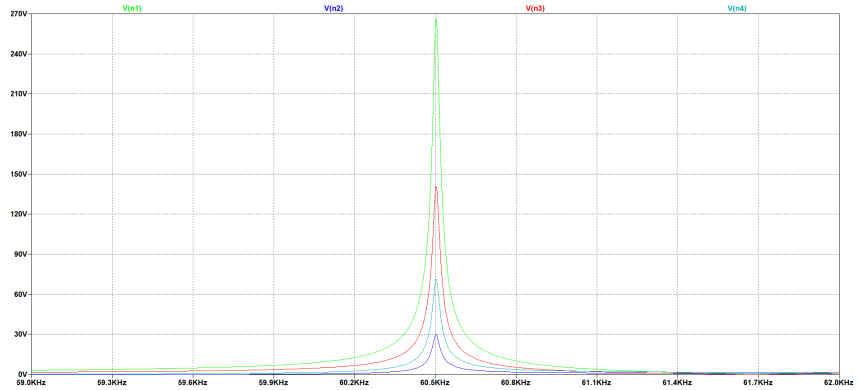


Figure: Node voltages on the left side of the chain ($N1$–$N4$).

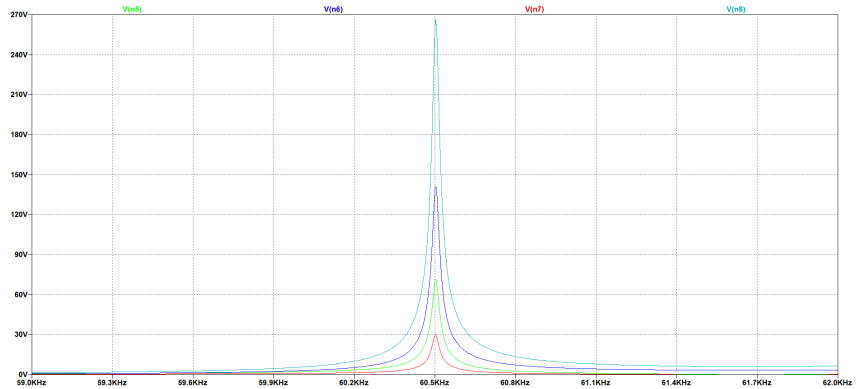


Figure: Node voltages on the right side of the chain ($N5$–$N8$).

LTSpice results: impedance resonance

The edge mode also shows up as a sharp peak in the two-point impedance, providing a clean spectral signature.

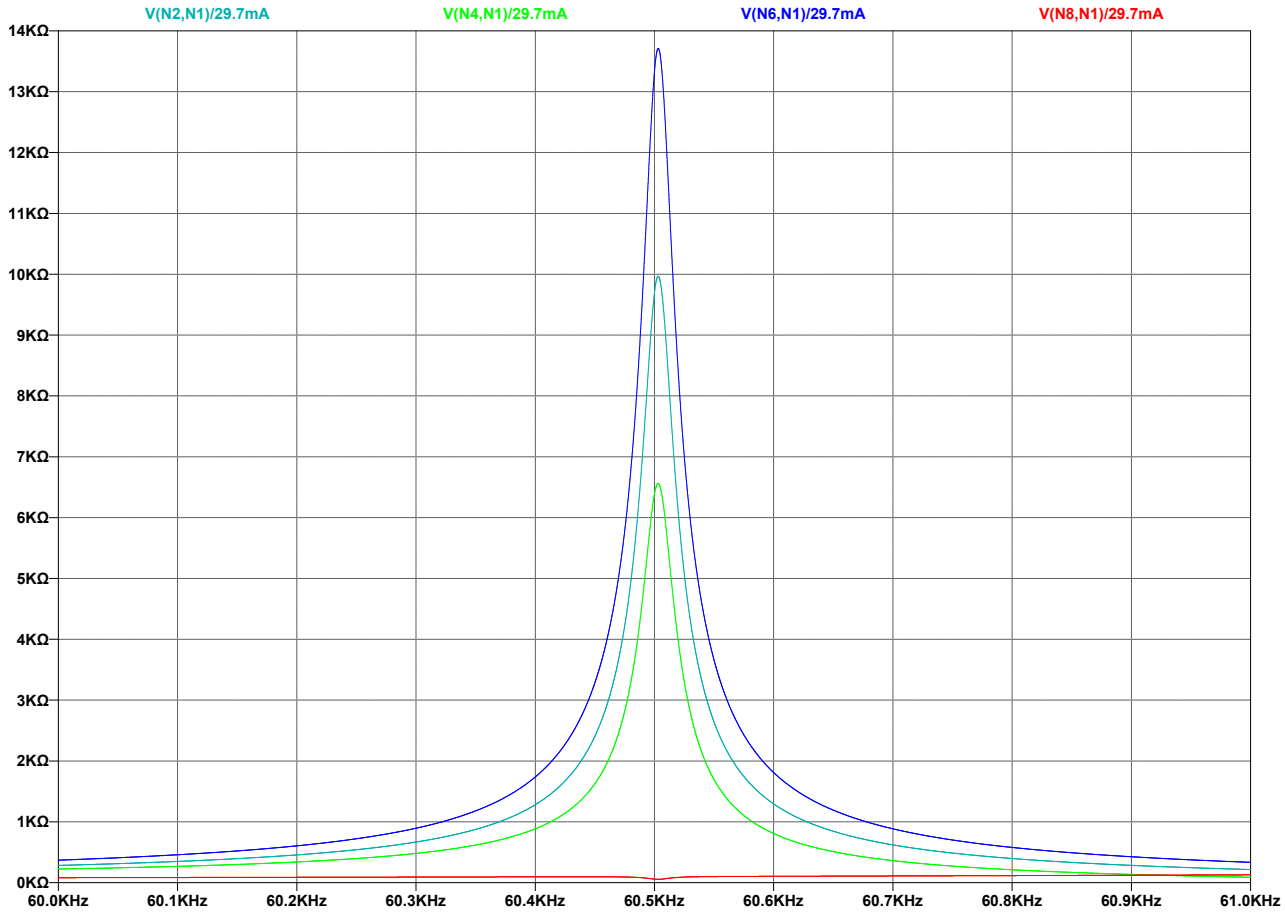


Figure: Two-point impedance $|Z_{k,1}(\omega)|$ showing a strong resonance peak.

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Hatano–Nelson model: non-reciprocal hopping ⇒ skin localization

The Hatano–Nelson (HN) chain is the simplest 1D model with direction-dependent coupling [4]:

$$H = \sum_n \left(t_R \, c_{n+1}^\dagger c_n \; + \; t_L \, c_n^\dagger c_{n+1} \right). \tag{18}$$

For open boundaries, non-reciprocity produces the **skin effect**: eigenmodes acquire an exponential envelope and pile up at one edge.

$$|\psi_n| \; \propto \; \left(\sqrt{\frac{|t_L|}{|t_R|}} \right)^n . \tag{19}$$

So $|t_R| > |t_L|$ accumulates modes towards one boundary, while $|t_L| > |t_R|$ accumulates them towards the other. This is the localization we will directly see in the circuit voltages.

[4] Naomichi Hatano and David R. Nelson. "Localization Transitions in Non-Hermitian Quantum Mechanics". In: *Phys. Rev. Lett.* 77 (3 July 1996), pp. 570–573. DOI: 10.1103/PhysRevLett.77.570

Circuit implementation: voltage follower creates one-way coupling

A voltage follower enforces one-way signal flow:

$$V_{\text{out}} = V_{\text{in}}. \tag{20}$$

Driving a link element from the follower output makes the downstream node "see" the upstream voltage, while the upstream node does not feel the downstream load. This makes the circuit Laplacian $J(\omega)$ **non-symmetric**, realizing $t_R \neq t_L$.

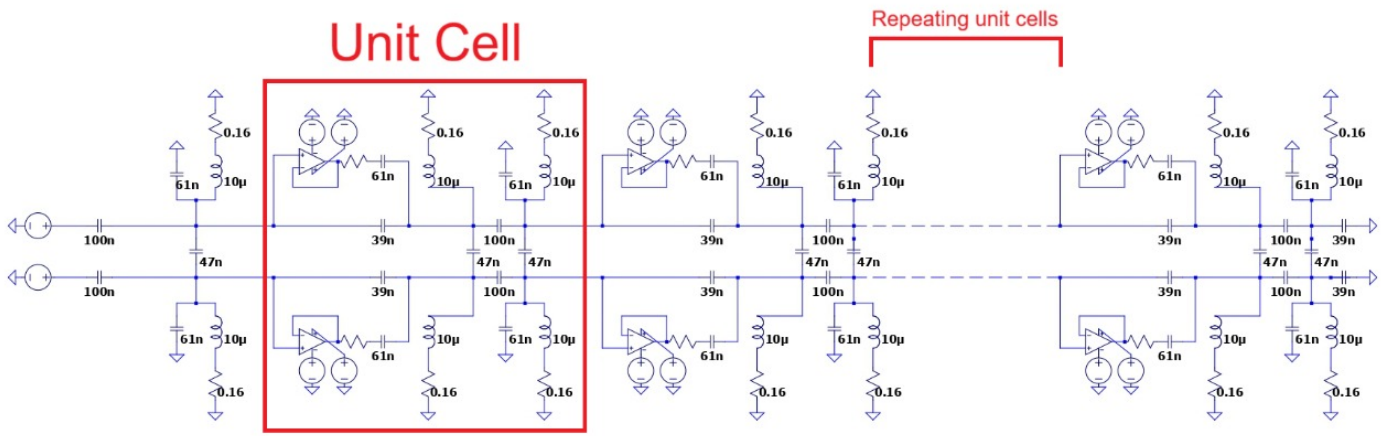


Figure: Unit-cell block (follower + passive elements) repeated to build a non-reciprocal SSH ladder.

LTSpice signature: boundary accumulation across modes

The skin effect shows up directly in the measured voltages: many resonant responses become biased toward one boundary when $t_R \neq t_L$.

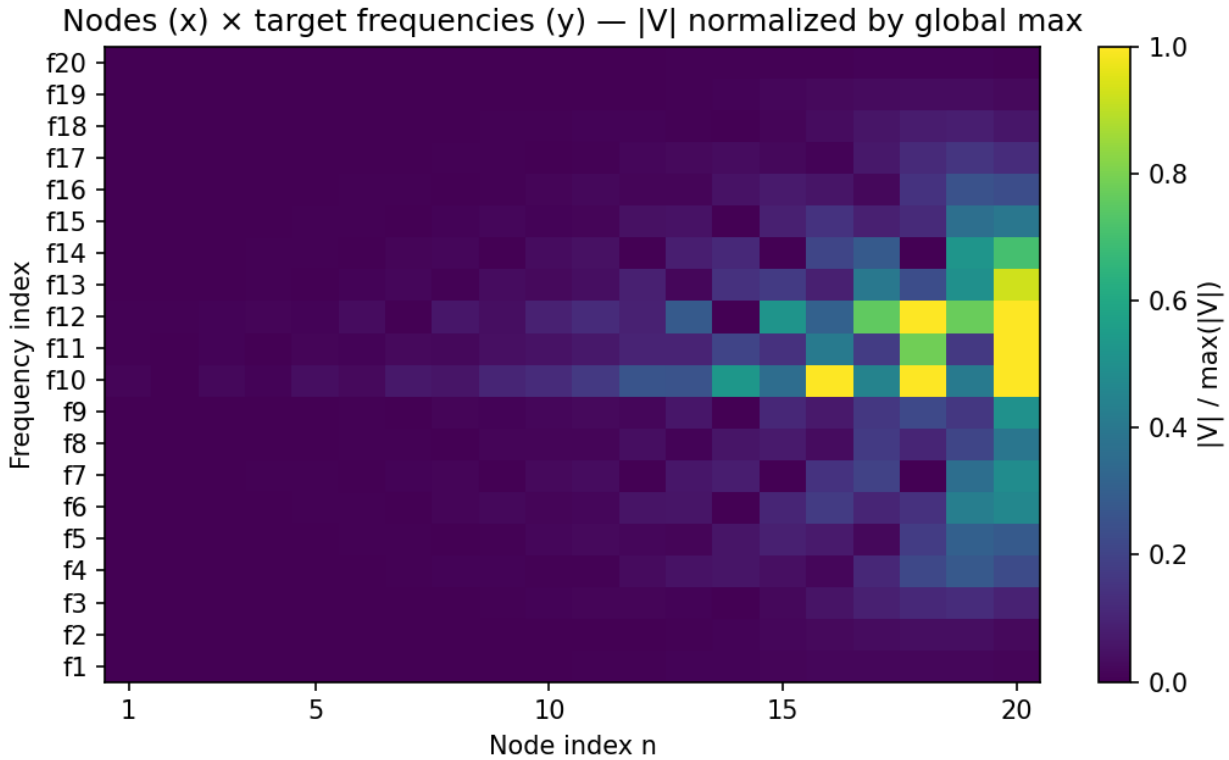


Figure: Frequency–node heatmap of normalized $|V_n|$ showing accumulation towards large n across multiple modes (skin effect).

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Where the field is going?

Topolectric circuits are now used as a lab-scale platform to explore lattice physics in regimes that are hard to realize in solids.

- **Unusual geometries and connectivities:** hyperbolic / non-Euclidean networks, fractal graphs, and designer lattices.
- **Scalable prototypes:** fast iteration by rewiring and swapping components, while keeping direct access to modes.

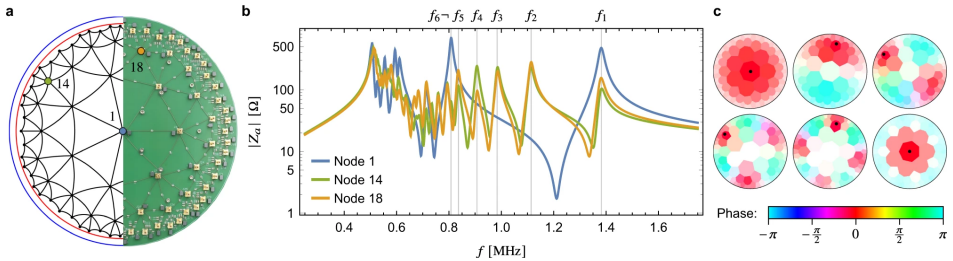


Figure: Example direction: circuits realizing hyperbolic (non-Euclidean) metamaterial lattices and probing modes via impedance/phase maps. Adapted from [5]

[5] Patrick M Lenggenhager et al. "Simulating hyperbolic space on a circuit board". In: *Nature Communications* 13.1 (July 2022), p. 4373

Complex phases and synthetic gauge fields in circuits

A major theme is engineering **complex couplings** (amplitude + phase), which enables synthetic gauge fields and richer topological phases.

- **Complex link weights:** circuit couplings behave like $t = a + ib = |t|e^{i\phi}$, giving controlled phases on links.
- **Flux and chirality:** phases around loops emulate magnetic flux, enabling chiral edge response and band topology.
- **Programmable networks:** active blocks and controlled sources make phases tunable in situ (and compatible with loss/gain).

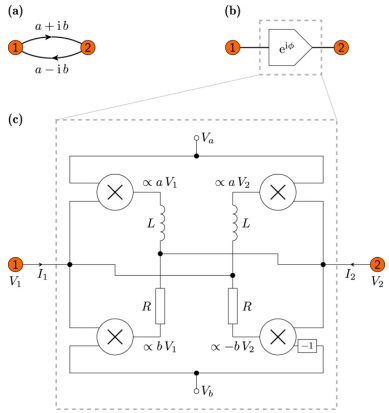


Figure: Complex coupling and a representative circuit block used to implement phase-controlled (effective) links.

Conclusion

Main message: topolectric circuits turn abstract lattice models into something you can *build*, *tune*, and *measure*.

- The grounded circuit Laplacian $J(\omega)$ is the core object: its eigenmodes are voltage patterns, and its inverse sets impedances.
- In the SSH circuit, topology becomes visible as a boundary-localized response together with a sharp impedance signature.
- In the Hatano–Nelson circuit, non-reciprocity pushes many modes to one edge (skin effect), again directly readable from voltages.
- Ongoing work in the field is expanding circuits into metamaterial geometries and phase-engineered couplings to access richer topological phases.

Thanks!

Questions and discussion welcome.

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- Hatano, Naomichi and David R. Nelson. "Localization Transitions in Non-Hermitian Quantum Mechanics". In: *Phys. Rev. Lett.* 77 (3 July 1996), pp. 570–573. DOI: 10.1103/PhysRevLett.77.570.
- Lee, Ching Hua et al. "Topolectrical Circuits". In: *Communications Physics* 1.1 (July 2018), p. 39.
- Batra, Navketan and Goutam Sheet. "Physics with Coffee and Doughnuts: Understanding the Physics Behind Topological Insulators Through Su-Schrieffer-Heeger Model". In: *Resonance* 25.6 (June 2020), pp. 765–786. ISSN: 0973-712X. DOI: 10.1007/s12045-020-0995-x.
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